

لا تنتسونا من صالح دعائكم / يعقوب الصلوي

If $y = f(x)$

(1)

بُرمز للمشتقة الاولى بأحد الرموز التالية

$\dot{y}$ or $\frac{dy}{dx}$ or y_1 or Dy

والمشتقة الثانية

$\ddot{y}$ or $\frac{d^2y}{dx^2}$ or y_2 or D^2y

والمشتقة الثالثة

$\dddot{y}$ or $\frac{d^3y}{dx^3}$ or y_3 or D^3y

والمشتقة الرابعة

$y^{(4)}$ or $\frac{d^4y}{dx^4}$ or y_4 or D^4y

⋮

والمشتقة النونية

$y^{(n)}$ or $\frac{d^ny}{dx^n}$ or y_n or D^ny

قواعد شهيرة في الإستقاق

① $y = k$, k constant $\rightarrow \frac{dy}{dx} = 0$
ثابت

Ex $y = 3 \rightarrow \frac{dy}{dx} = 0$, $y = \frac{\pi}{2} \rightarrow \frac{dy}{dx} = 0$
ثابت

② $y = ax$ $\rightarrow \frac{dy}{dx} = a$
ثابت

Ex $y = 3x \rightarrow \frac{dy}{dx} = 3$, $y = x \rightarrow \frac{dy}{dx} = 1$

③ $y = x^n \rightarrow \frac{dy}{dx} = nx^{n-1}$
Ex $y = x^3 \rightarrow \frac{dy}{dx} = 3x^2$

$$(1) y = x^{-3}$$

$$(2) y = x^{-5}$$

$$(3) y = x^{-1}$$

أمثلة مشابهة

$$(4) y = (1+x)^{-1}$$

$$(5) y = (1+x)^{-3}$$

$$(6) y = (1-x)^{-5}$$

الحل

$$(1) y = x^{-3} \rightarrow y' = -3 x^{-3-1} = -3 x^{-4}$$

$$(2) y = x^{-5} \rightarrow y' = -5 x^{-5-1} = -5 x^{-6}$$

$$(3) y = x^{-1} \rightarrow y' = -1 x^{-1-1} = -x^{-2}$$

$$(4) y = (1+x)^{-1} \rightarrow y' = (-1) (1+x)^{-2} (1) = - (1+x)^{-2}$$

$$(5) y = (1+x)^{-3} \rightarrow y' = -3 (1+x)^{-4} (1) = -3 (1+x)^{-4}$$

$$(6) y = (1-x)^{-5} \rightarrow y' = -5 (1-x)^{-6} (-1) = 5 (1-x)^{-6}$$

أمثلة مشابهة

④ $y = (f(x))^n \rightarrow \frac{dy}{dx} = n (f(x))^{n-1} \cdot f'(x)$ (2)

مشتقة ما بداخل القوس

Ex $y = (x^2 + 1)^7 \rightarrow \frac{dy}{dx} = 7 (x^2 + 1)^6 \cdot (2x) = 14x (x^2 + 1)^6$

$y = (\pi - x)^2 \rightarrow \frac{dy}{dx} = 2(\pi - x)(-1) = -2(\pi - x)$

تابع لاحقاً أمثلة أخرى

مشتقة ما بداخل القوس (ثابت)

⑤ $y = \sqrt{f(x)} \rightarrow \frac{dy}{dx} = \frac{f'(x)}{2\sqrt{f(x)}}$

مشتقة ما تحت الجذر
× 2

Ex ① $y = \sqrt{x} \rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

② $y = \sqrt{1 - x^2} \rightarrow \frac{dy}{dx} = \frac{-2x}{2\sqrt{1 - x^2}} = \frac{-x}{\sqrt{1 - x^2}}$

(8) (ملاحظة) جداً جداً ...

من أجل
السوية

حالات خاصة
مهمة جداً جداً

$$\textcircled{1} \left(\frac{a}{x^n} \right)' = \frac{-a(n)}{x^{n+1}}$$

EX

الشرط البسيط
ثابت

$$\frac{\text{سالب العدد} \times \text{القوة}}{1 + \text{القوة}} = \left(\frac{\text{عدد}}{x^{\text{قوة}}} \right)'$$

$$y = \frac{3}{x^2} \rightarrow \frac{dy}{dx} = \frac{-(3)(2)}{x^{2+1}} = \frac{-6}{x^3}$$

$$y = \frac{-2}{x^5} \rightarrow \frac{dy}{dx} = \frac{-(-2)(5)}{x^{5+1}} = \frac{10}{x^6}$$

$$y = \frac{1}{x} \rightarrow \frac{dy}{dx} = \frac{-1}{x^2}$$

$$y = \frac{-1}{x} \rightarrow \frac{dy}{dx} = \frac{1}{x^2}$$

$$y = \frac{-3}{x^2} \rightarrow \frac{dy}{dx} = \frac{6}{x^3}$$

$$\textcircled{2} \left[\frac{a}{[f(x)]^n} \right]' = \frac{-a(n) \cdot f'(x)}{(f(x))^{n+1}}$$

الشرط البسيط
ثابت

ما هو
الشرط البسيط
نؤكد إشارة البسيط $\times$ القوة $\times$ مشتق القوة

$$y = \frac{3}{(x^2+1)^4} \rightarrow \frac{dy}{dx} = \frac{(-3)(4)(2x)}{(x^2+1)^{4+1}} = \frac{-24x}{(x^2+1)^5}$$

$$y = \frac{2}{(1+x)^3} \rightarrow \frac{dy}{dx} = \frac{-2(3)(1)}{(1+x)^4} = \frac{-6}{(1+x)^4}$$

$$y = \frac{1}{(1-x)^2} \rightarrow \frac{dy}{dx} = \frac{-1(2)(-1)}{(1-x)^3} = \frac{2}{(1-x)^3}$$

$$y = \frac{-1}{1-2x} \rightarrow \frac{dy}{dx} = \frac{-(-1)(1)(-2)}{(1-2x)^2} = \frac{-2}{(1-2x)^2}$$

ويمكن حل
بجهد نضع لقوى
الى البسيط باشارة مخالفة للاشارة

لكن هذا ليس شرط البسيط
بشرط البسيط
ومشتق البسيط

$$(7) \quad y = \log_{(e)}(f(x)) \rightarrow \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

ملحقة ما بعد اللوغاريتم
ما بعد اللوغاريتم نفسه

$$(3) \quad \log_e(x) = \ln(x)$$

$$y = \log(x) \rightarrow \frac{dy}{dx} = \frac{1}{x}$$

$$y = \ln(x) \rightarrow \frac{dy}{dx} = \frac{1}{x}$$

$$y = \log(x^2 + 1) \rightarrow \frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

$$y = \log(1 - 2x) \rightarrow \frac{dy}{dx} = \frac{-2}{1 - 2x}$$

ملاحظات للتبسيط في اللوغاريتمات

$$(1) \quad \log\left(\frac{A}{B}\right) = \log(A) - \log(B)$$

$$(2) \quad \log(A \cdot B) = \log(A) + \log(B)$$

$$(3) \quad \log(A)^m = m \cdot \log(A)$$

$$(4) \quad \log\left(\frac{A \cdot B}{C \cdot D}\right) = \log(A) + \log(B) - \log(C) - \log(D)$$

$$\log(1) = 0$$

$$\ln(e) = 1$$

$$\ln(f(x)) = f(x)$$

ثابت a

$$(6) \quad y = a^{f(x)} \rightarrow \frac{dy}{dx} = a^{f(x)} \cdot f'(x) \cdot \log(a)$$

$\frac{dy}{dx}$ الرتبة نفسها
 a الأساس
 $f'(x)$ مشتقة
 $\log(a)$ لوغاريتم الأساس

$$y = e^{f(x)} \rightarrow \frac{dy}{dx} = e^{f(x)} \cdot f'(x)$$

EX (1) $y = 3^{2x} \rightarrow \frac{dy}{dx} = 3^{2x} \cdot (2) \cdot \log(3)$

(2) $y = 2^{x^3+1} \rightarrow \frac{dy}{dx} = 2^{x^3+1} \cdot (3x^2) \cdot \log(2)$

(3) $y = e^{5x} \rightarrow \frac{dy}{dx} = e^{5x} \cdot (5) = 5e^{5x}$

(4) $y = e^x \rightarrow \frac{dy}{dx} = e^x (1) = e^x$

(5) $y = e^{-x} \rightarrow \frac{dy}{dx} = e^{-x} (-1) = -e^{-x}$

$\sin \rightarrow$ جا

$\cos \rightarrow$ قبا

$\tan \rightarrow$ ظا

$\cot \rightarrow$ قظا

$\sec \rightarrow$ قفا

$\csc \rightarrow$ قنفا

$\boxed{\text{CSC}}$

قواعد التبسيط

$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$

$\sec(\theta) = \frac{1}{\cos(\theta)}$

$\csc(\theta) = \frac{1}{\sin(\theta)}$

$\cot(\theta) = \frac{1}{\tan(\theta)}$

$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$

30°

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$

45°

$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

$\sin(0) = 0$

$\sin(\pi) = 0$

$\cos(0) = 1$

$\cos(\pi) = -1$



$\sin\left(\frac{\pi}{2}\right) = 1$ $\cos\left(\frac{\pi}{2}\right) = 0$

$\sin\left(\frac{3\pi}{2}\right) = -1$ $\cos\left(\frac{3\pi}{2}\right) = 0$

قواعد الاشتقاق

$\left(\sin(f(x))\right)' = \cos(f(x)) \cdot f'(x)$

$\left(\cos(f(x))\right)' = -\sin(f(x)) \cdot f'(x)$

$\left(\tan(f(x))\right)' = \sec^2(f(x)) \cdot f'(x)$

$\left(\cot(f(x))\right)' = -\csc^2(f(x)) \cdot f'(x)$

$\left(\sec(f(x))\right)' = \sec(f(x)) \tan(f(x)) \cdot f'(x)$

$\left(\csc(f(x))\right)' = -\csc(f(x)) \cot(f(x)) \cdot f'(x)$

ملاحظة

$\tan$ $\sec$
 $\cot$ $\csc$

المثال ٢

$$y = \frac{a}{(f(x))^n} \rightarrow y' = \frac{-a(n) f'(x)}{(f(x))^{n+1}}$$

ملاحظة: a عدد، $f(x)$ دالة، n أس.

$$① y = \frac{3}{(x^2+1)^5} \rightarrow y' = \frac{-3(5)(2x)}{(x^2+1)^6} = \frac{-30x}{(x^2+1)^6}$$

$$② y = \frac{-2}{(3x+1)^7} \rightarrow y' = \frac{(-2)(7)(3)}{(3x+1)^8} = \frac{-42}{(3x+1)^8}$$

$$③ y = \frac{1}{(1+x)^2} \rightarrow y' = \frac{-(1)(2)(1)}{(1+x)^3} = \frac{-2}{(1+x)^3}$$

$$④ y = \frac{1}{1+x} \rightarrow y' = \frac{-(1)(1)(1)}{(1+x)^2} = \frac{-1}{(1+x)^2}$$

$$y = \sin(2x) \rightarrow y' = \cos(2x)(2) = 2\cos(2x)$$

$$y = \cos(3x) \rightarrow y' = -\sin(3x)(3) = -3\sin(3x)$$

$$y = 2\sin(4x) \rightarrow y' = 2\cos(4x)(4) = 8\cos(4x)$$

$$y = \sin^2(x) \Rightarrow y = (\sin x)^2 \rightarrow y' = 2(\sin x)' \cos(x)$$

قانون
 $\sin(2x) = 2\sin(x)\cos(x)$

$$= 2\sin(x)\cos(x)$$

$$= \sin(2x)$$

$$y = \cos^2(x) \rightarrow y = (\cos x)^2 \rightarrow y' = 2(\cos x)'(-\sin x)$$

$$= -2\sin(x)\cos(x) = -\sin(2x)$$

(10) مشتقة لرفال الزائدية

$$\left(\sinh(f(x)) \right)' = \cosh(f(x)) \cdot f'(x)$$

$$\left(\cosh(f(x)) \right)' = \sinh(f(x)) \cdot f'(x)$$

$$\left(\tanh(f(x)) \right)' = \operatorname{sech}^2(f(x)) \cdot f'(x)$$

$$\left(\coth(f(x)) \right)' = -\operatorname{cosech}^2(f(x)) \cdot f'(x)$$

$$\left(\operatorname{sech}(f(x)) \right)' = -\operatorname{sech}(f(x)) \cdot \tanh(f(x)) \cdot f'(x)$$

$$\left(\operatorname{cosech}(f(x)) \right)' = -\operatorname{cosech}(f(x)) \cdot \coth(f(x)) \cdot f'(x)$$

اشتقاقيا

نفس النسب مثلثية فقط الثلاث الاولى
مشتقها موجب والثلاث الاخيرة مشتقها سالب

اوجد ثلاث مشتقات للدوال التالية

① $y = e^x$

② $y = 3^x$

③ $y = a^x$

④ $y = \sin(x)$

⑤ $y = \cos(x)$

⑥ $y = \sin(2x)$

⑦ $y = \cos^2(x)$

⑧ $y = \log(x)$

⑨ $y = \frac{1}{x} \rightarrow$ ⑩ $y = \frac{1}{1+x}$

⑪ $y = \log(1+x)$

⑫ $y = \ln(1-x)$

⑬ $y = \sin(x) + \cos(x)$

⑭ $y = \sin\left(\frac{x}{2}\right)$

⑮ $y = \cos\left(\frac{x}{2}\right)$

⑯ $y = \frac{1}{1+2x}$

⑰ $y = \log(1+2x)$

⑱ $y = \log(\cos x)$

⑳ $y = \frac{1}{\sqrt{x}}$

㉑ $y = \sin^2(x)$

㉒ $y = (1+x)^4$

㉓ $y = (1+x)^n$

㉔ $y = \log(\sec x)$

حل تمارين المرحلة الثانية

① $y = x e^x - \log(1+x)$

حاصل ضرب والتبني

$$y' = x(e^x)' + e^x(x)' - \frac{1}{1+x}$$

$$y' = x e^x + e^x(1) - \frac{1}{1+x}$$

حاصل ضرب

$$y' = x e^x + e^x(1) + e^x - \frac{1}{(1+x)^2}$$

$$y' = x e^x + 2e^x + \frac{1}{(1+x)^2} \quad (T)$$

② $y = x \cos(x) - \sin(x)$

حاصل ضرب

$$y' = x(\cos x)' + \cos(x)(x)' - \cos x$$

$$y' = -x \sin x + \cos(x)(1) - \cos(x)$$

$$y' = -x \sin x$$

حاصل ضرب

$$y' = (-x)(\sin x)' + \sin(x)(-x)'$$

$$y' = -x \cos(x) + \sin(x)(-1)$$

$$y' = -x \cos(x) - \sin(x) \quad (T)$$

③ $y = \log(a^x + b^x + c^x) - \log(3)$

مشتقة لاسي

ثابت

مشتقة من

$$y' = \frac{a^x(1)\log(a) + b^x(1)\log(b) + c^x(1)\log(c)}{a^x + b^x + c^x} - 0$$

$$= \frac{a^x \log(a) + b^x \log(b) + c^x \log(c)}{a^x + b^x + c^x} \quad (T)$$

تذكيران

قانون شرط البسط عدد

$$\left(\frac{a}{(f(x))^n} \right)' = \frac{-a(n) \cdot f'(x)}{(f(x))^{n+1}}$$

$$\left(\frac{1}{(1+x)^1} \right)' = \frac{-1(1)(1)}{(1+x)^2}$$

$$= \frac{-1}{(1+x)^2}$$

ويمكن اشتقاق كدالة كسرية

$$\left(\frac{1}{1+x} \right)' = \frac{(1)'(1+x) - (1)(1)'}{(1+x)^2}$$

$$= \frac{(0)(1+x) - (1)(1)'}{(1+x)^2}$$

$$= \frac{-1}{(1+x)^2}$$

زيادة مضاعفة

at $x=0$

$$y(0)$$

$$= \frac{a^0 \log(a) + b^0 \log(b) + c^0 \log(c)}{a^0 + b^0 + c^0}$$

$$= \frac{1 \log(a) + 1 \log(b) + 1 \log(c)}{1 + 1 + 1}$$

$$= \frac{\log(a \cdot b \cdot c)}{3}$$

$$= \frac{1}{3} \log(a \cdot b \cdot c)$$

$$= \log \sqrt[3]{abc}$$

⑦ $y = (e^x - 1) \sin x$
 حاصل ضرب دالتين

$$\dot{y} = (e^x - 1)' \sin(x) + (\sin x)' (e^x - 1)$$

$$= (e^x - 0) \sin(x) + \cos(x) (e^x - 1)$$

$$\dot{y} = e^x \sin x + e^x \cos(x) - \cos(x)$$

حاصل ضرب حاصل ضرب

$$\ddot{y} = (e^x)' \sin x + (\sin x)' e^x + (e^x)' \cos(x) + (\cos x)' e^x - (-\sin x)$$

$$\ddot{y} = e^x \sin x + e^x \cos(x) + e^x \cos(x) - e^x \sin x + \sin x$$

$$\ddot{y} = 2e^x \cos(x) + \sin(x)$$

(T)

⑧ $y = e^x - e^{-x} - 2 \log(1+x)$ $\dot{y}(0)$

$$\dot{y} = e^x(1) - e^{-x}(-1) - 2 \cdot \frac{1}{1+x}$$

$$\dot{y} = e^x + e^{-x} - \frac{2}{1+x}$$

$$\ddot{y} = e^x(1) + e^{-x}(-1) - \frac{-2(1)(1)}{(1+x)^2}$$

$$\ddot{y} = e^x - e^{-x} + \frac{2}{(1+x)^2}$$

$$\ddot{y}(0) = e^0 - e^0 + \frac{2}{(1+0)^2} = 1 - 1 + \frac{2}{1} = 0 + 2 = 2$$

(T)

⑨ $y = (x - \frac{\pi}{2}) \sin x \rightarrow \dot{y} = (x - \frac{\pi}{2})' \sin(x) + (\sin(x))' (x - \frac{\pi}{2})$
 حاصل ضرب دالتين ثابت مشتقة صفر

$$\dot{y} = (1) \sin(x) + (x - \frac{\pi}{2}) \cos(x)$$

when $x = \frac{\pi}{2}$ $\dot{y}(\frac{\pi}{2}) = \sin(\frac{\pi}{2}) + (\frac{\pi}{2} - \frac{\pi}{2}) \cos(\frac{\pi}{2}) = 1 + 0 = 1$ (T)

نحل

$$\sin(\frac{\pi}{2}) = 1$$

$$\cos(\frac{\pi}{2}) = 0$$

حل التمارين

(1) $y = e^x$
 $y' = e^x$ (1) $= e^x$
 $y'' = e^x$
 $y''' = e^x$

(2) $y = 3^x$
 $y' = 3^x \log(3)$
 $y'' = 3^x (\log(3))^2$
 $y''' = 3^x (\log(3))^3$

(3) $y = a^x$
 $y' = a^x \log(a)$
 $y'' = a^x (\log(a))^2$
 $y''' = a^x (\log(a))^3$

(4) $y = \sin(x)$
 $y' = \cos(x)$
 $y'' = -\sin(x)$
 $y''' = -\cos(x)$

(5) $y = \cos(x)$
 $y' = -\sin(x)$
 $y'' = -\cos(x)$
 $y''' = -(-\sin(x)) = \sin(x)$

(6) $y = \sin(2x)$
 $y' = \cos(2x) (2) = 2\cos(2x)$
 $y'' = 2(-\sin(2x)) (2) = -4\sin(2x)$
 $y''' = -4\cos(2x) (2) = -8\cos(2x)$

(7) $y = \cos^2(x)$
 $y' = 2(\cos(x))(-\sin(x)) = -2\sin(x)\cos(x) = -\sin(2x)$
 $y'' = -\cos(2x) (2) = -2\cos(2x)$
 $y''' = -2(-\sin(2x)) (2) = 4\sin(2x)$

(8) $y = \log(x)$

$y' = \frac{1}{x}$
 $y'' = -\frac{1}{x^2}$
 $y''' = \frac{2}{x^3}$

(9) $y = \frac{1}{x}$

$y' = -\frac{1}{x^2}$
 $y'' = \frac{2}{x^3}$
 $y''' = -\frac{6}{x^4}$

تعليم ان
 $\sin(2x) = 2\sin(x)\cos(x)$

قانون
 $\left(\frac{a}{x^n}\right)' = -\frac{a(n)}{x^{n+1}}$

$$(10) \quad y = \frac{1}{1+x} \rightarrow \frac{1}{(1+x)^1}$$

$$\dot{y} = \frac{-1(1)}{(1+x)^2} = \frac{-1}{(1+x)^2}$$

$$\ddot{y} = \frac{2(1)}{(1+x)^3} = \frac{2}{(1+x)^3}$$

$$\dddot{y} = \frac{-6(1)}{(1+x)^4} = \frac{-6}{(1+x)^4}$$

$$(11) \quad y = \log(1+x)$$

$$\dot{y} = \frac{1}{1+x} \rightarrow \frac{1}{(1+x)^1}$$

$$\ddot{y} = \frac{-1(1)}{(1+x)^2} = \frac{-1}{(1+x)^2}$$

$$\ddot{y} = \frac{2(1)}{(1+x)^3} = \frac{2}{(1+x)^3}$$

$$(12) \quad y = \ln(1-x)$$

$$\dot{y} = \frac{-1}{1-x} \rightarrow \frac{-1}{(1-x)^1}$$

$$\ddot{y} = \frac{1(-1)}{(1-x)^2} = \frac{-1}{(1-x)^2}$$

$$\ddot{y} = \frac{2(-1)}{(1-x)^3} = \frac{-2}{(1-x)^3}$$

$$(13) \quad y = \sin(x) + \cos(x)$$

$$\dot{y} = \cos(x) - \sin(x)$$

$$\ddot{y} = -\sin(x) - \cos(x)$$

$$\ddot{y} = -\cos(x) - (-\sin(x))$$

$$= -\cos(x) + \sin(x)$$

$$(14) \quad y = \sin\left(\frac{x}{2}\right)$$

$$\dot{y} = \cos\left(\frac{x}{2}\right) \left(\frac{1}{2}\right)$$

$$= \frac{1}{2} \cos\left(\frac{x}{2}\right)$$

$$\ddot{y} = \frac{1}{2} (-\sin\left(\frac{x}{2}\right)) \left(\frac{1}{2}\right) = -\frac{1}{4} \sin\left(\frac{x}{2}\right)$$

$$\ddot{y} = -\frac{1}{4} \cos\left(\frac{x}{2}\right) \left(\frac{1}{2}\right) = -\frac{1}{8} \cos\left(\frac{x}{2}\right)$$

$$(15) \quad y = \cos\left(\frac{x}{2}\right)$$

$$\dot{y} = -\sin\left(\frac{x}{2}\right) \left(\frac{1}{2}\right) = -\frac{1}{2} \sin\left(\frac{x}{2}\right)$$

$$\ddot{y} = -\frac{1}{2} \cos\left(\frac{x}{2}\right) \left(\frac{1}{2}\right) = -\frac{1}{4} \cos\left(\frac{x}{2}\right)$$

$$\ddot{y} = -\frac{1}{4} (-\sin\left(\frac{x}{2}\right)) \left(\frac{1}{2}\right) = \frac{1}{8} \sin\left(\frac{x}{2}\right)$$

$$(16) \quad y = \frac{1}{1+2x} \rightarrow \frac{1}{(1+2x)^1}$$

$$\dot{y} = \frac{-1(2)}{(1+2x)^2} = \frac{-2}{(1+2x)^2}$$

$$\ddot{y} = \frac{4(2)}{(1+2x)^3} = \frac{8}{(1+2x)^3}$$

$$\ddot{y} = \frac{-24(2)}{(1+2x)^4} = \frac{-48}{(1+2x)^4}$$

$$(17) \quad y = \log(1+2x)$$

$$\dot{y} = \frac{2}{1+2x} \rightarrow \frac{2}{(1+2x)^1}$$

$$\ddot{y} = \frac{-2(2)}{(1+2x)^2} = \frac{-4}{(1+2x)^2}$$

$$\ddot{y} = \frac{8(2)}{(1+2x)^3} = \frac{16}{(1+2x)^3}$$

⑤

ملخص ما سبق
من القواعد

$$\textcircled{1} \quad \sin^2(x) + \cos^2(x) = 1 \Rightarrow \begin{cases} \sin^2(x) = 1 - \cos^2(x) \\ \cos^2(x) = 1 - \sin^2(x) \end{cases}$$

$$\textcircled{2} \quad 1 + \tan^2(x) = \sec^2(x)$$

$$\textcircled{3} \quad 1 + \cot^2(x) = \operatorname{cosec}^2(x)$$

تذكر أيضا

$$\tan(x) = \frac{\sin x}{\cos x}$$

$$\cot(x) = \frac{1}{\tan(x)}$$

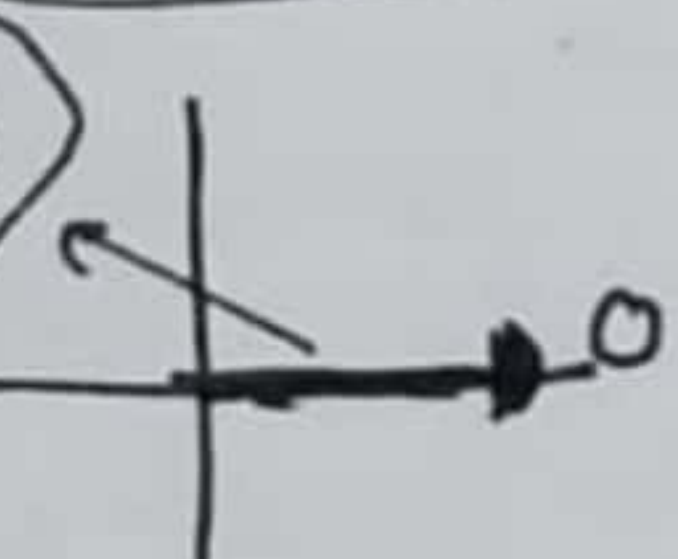
$$\sec(x) = \frac{1}{\cos(x)}$$

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

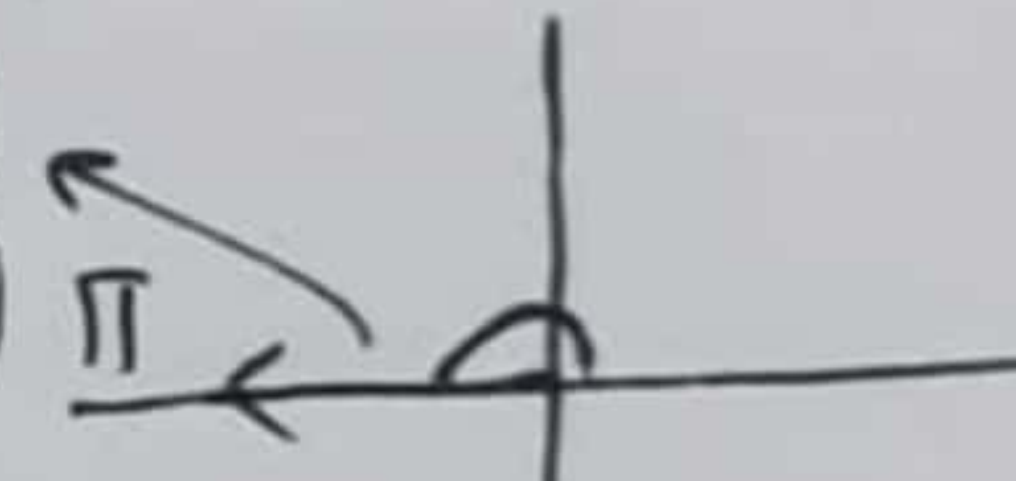
$$\operatorname{cosec}(x) = \frac{1}{\sin(x)}$$

قيم لنسب المثلثية لزوايا التورية المحورية

الزاوية صفر منطبق على محور
x موجب
يعني الضلع المقابل
= صفر



الزاوية π منطبق على
محور x سالب
أيضا المقابل = صفر



$$\sin(\pi) = 0 \\ \cos(\pi) = -1$$

$$\sin(0) = 0 \\ \cos(0) = 1$$

$$\frac{0}{1} = 0 \\ \frac{1}{0} = \infty$$

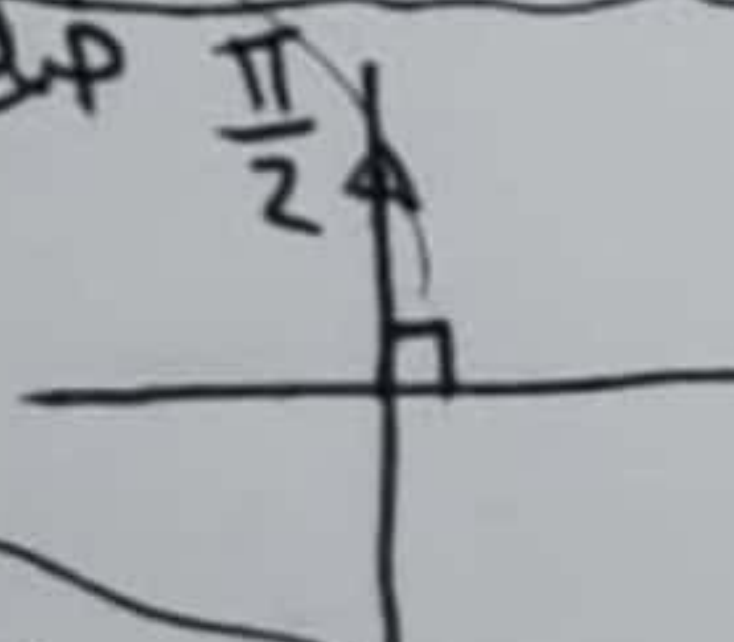
$$\tan(0) = 0 \rightarrow \textcircled{\text{لأن}} \tan(0) = \frac{\sin(0)}{\cos(0)} = \frac{0}{1} = 0$$

$$\sec(0) = 1 \rightarrow \textcircled{\text{لأن}} \sec(0) = \frac{1}{\cos(0)} = \frac{1}{1} = 1$$

$$\cot(0) = \frac{\cos(0)}{\sin(0)} = \frac{1}{0} = \infty \quad \operatorname{cosec}(0) = \frac{1}{\sin(0)} = \frac{1}{0} = \infty$$

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الزاوية $\frac{\pi}{2}$ منطبق على
محور y الموجب
يعني الضلع المجاور
= صفر



$$\cos\left(\frac{\pi}{2}\right) = 0$$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

$$\tan\left(\frac{\pi}{2}\right) = \frac{\sin \frac{\pi}{2}}{\cos \frac{\pi}{2}} = \frac{1}{0} = \infty$$

$$\cot\left(\frac{\pi}{2}\right) = \frac{\cos \frac{\pi}{2}}{\sin \frac{\pi}{2}} = \frac{0}{1} = 0$$

محور y السالب

$$\cos\left(\frac{3\pi}{2}\right) = 0$$

$$\sin\left(\frac{3\pi}{2}\right) = -1$$

تذكير سريع

ملخص الحصة

8

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\sec(x) = \frac{1}{\cos(x)}$$

$$\csc(x) = \frac{1}{\sin(x)}$$

$$\cot(x) = \frac{1}{\tan(x)}$$

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$\sin^2(x) + \cos^2(x) = 1$$

$$\sin^2(x) = 1 - \cos^2(x)$$

$$\cos^2(x) = 1 - \sin^2(x)$$

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$1 - \cos(x) = 2 \sin^2\left(\frac{x}{2}\right)$$

$$1 + \cos(x) = 2 \cos^2\left(\frac{x}{2}\right)$$

$$\sin(0) = 0$$

$$\sin(\pi) = 0$$

$$\cos(0) = 1$$

$$\cos(\pi) = -1$$

π | 0

زاوية على محور السينات
قيمة $\sin$ صفر

$\frac{\pi}{2}$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

$$\cos\left(\frac{\pi}{2}\right) = 0$$

$$\sin\left(\frac{3\pi}{2}\right) = -1$$

$$\cos\left(\frac{3\pi}{2}\right) = 0$$

$\frac{3\pi}{2}$

زاوية على محور الصادات
قيمة $\cos$ صفر

$$\tan(0) = 0$$

$$\sec(0) = 1$$

توضيح

$$\tan(0) = \frac{\sin 0}{\cos 0} = \frac{0}{1} = 0$$

$$\sec(0) = \frac{1}{\cos(0)} = \frac{1}{1} = 1$$

$$\frac{0}{3} = 0$$

صفر = صفر
عدد ≠ صفر

$$\frac{3}{\infty} = 0$$

عدد = صفر
عدد

$$3 \times 0 = 0$$

عدد $\times$ صفر = صفر

العكس
 $\frac{1}{0} = \infty$ $\infty = \frac{\text{عدد}}{\text{صفر}}$

$$\log(1) = 0$$

$$\ln(1) = 0$$

$$\frac{0}{3} = 1$$

$$e^0 = 1$$

(عدد) صفر = 1
العدد ط 0

$$e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

$$\log(0) = -\infty$$

$$e^{-\infty} = 0$$

$$\log(\infty) = \infty$$

$$e^{\infty} = \infty$$

$$\frac{\infty}{3} = \infty$$

$$\log(1) = 0$$

$$\ln(e) = 1$$

$$\log(\infty) = \infty$$

$$\log(0) = -\infty$$

$$e^0 = 1$$

$$e^{\infty} = \infty$$

خواص

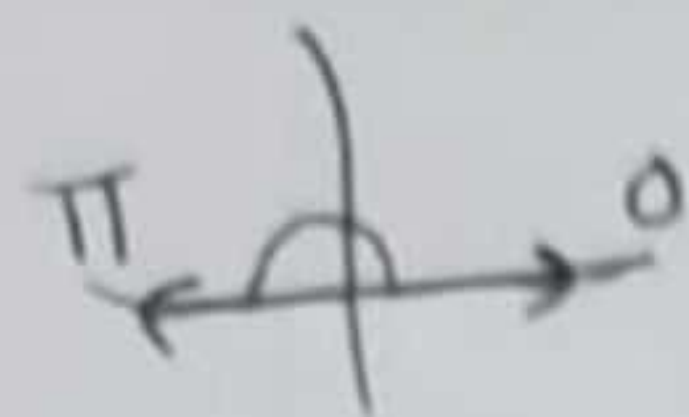
$$\log\left(\frac{A}{B}\right) = \log(A) - \log(B)$$

$$\log(A \times B) = \log(A) + \log(B)$$

$$\log(A^m) = m \log(A)$$

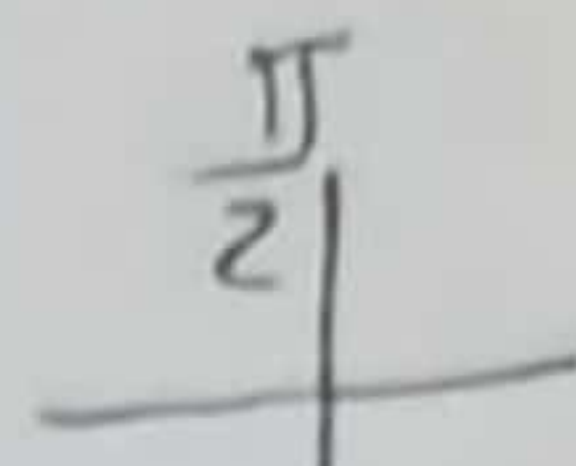
6

ملخص ما سبق
للثلاث



$$\sin(0) = 0 \quad \sin(\pi) = 0$$

$$\cos(0) = 1 \quad \cos(\pi) = -1$$



$$\sin\left(\frac{\pi}{2}\right) = 1$$

$$\cos\left(\frac{\pi}{2}\right) = 0$$

لايجاد tan / sec
وبقية النسب

نستخدم القواعد

$$\tan(\) = \frac{\sin(\)}{\cos(\)}$$

$$\cot(\) = \frac{\cos(\)}{\sin(\)}$$

$$\sec(\) = \frac{1}{\cos(\)}$$

$$\operatorname{cosec}(\) = \frac{1}{\sin(\)}$$

$$\frac{0}{1} = 0$$

$$\frac{1}{0} = \infty$$

ملاحظات في الكسور

$$\textcircled{1} \quad \frac{\frac{a}{b}}{\frac{c}{m}} = \frac{a \times m}{b \times c}$$

$$\textcircled{2} \quad \frac{\frac{a}{b}}{c} = \frac{a}{b \times c}$$

$$\textcircled{3} \quad \frac{a}{\frac{b}{c}} = \frac{a \times c}{b}$$

بسط ما يلي

$$\textcircled{1} \quad \frac{\frac{x}{\sin x}}{\frac{1}{\cos x}} = \frac{x \cos x}{\sin x}$$

$$\textcircled{2} \quad \frac{x}{\sec(x)} = \frac{x}{\frac{1}{\cos x}} = x \cos x$$

$$\begin{aligned} \textcircled{3} \quad \frac{\operatorname{cosec}^2 x}{\cot x} &= \frac{\operatorname{cosec}^2 x}{\cot x} = \frac{\operatorname{cosec} x}{\cot^2 x} = \frac{\frac{1}{\sin x}}{\frac{\cos^2 x}{\sin^2 x}} \\ &= \frac{\sin^2 x}{\sin x \cos^2 x} = \frac{\sin x}{\cos^2 x} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad \frac{\frac{2x}{1+x}}{4x} &= \frac{2x}{4x(1+x)} \\ &= \frac{1}{2(1+x)} \end{aligned}$$

If $y = e^x \sin(x)$ then $\ddot{y} = \dots$

حاصل ضرب دالتين

$$\dot{y} = (e^x)' \sin(x) + (\sin x)' e^x$$

$$\dot{y} = e^x \sin(x) + e^x \cos(x)$$

حاصل ضرب حاصل ضرب

$$\ddot{y} = e^x \sin(x) + e^x \cos(x) + e^x \cos(x) + e^x (-\sin x)$$

متساوية نجعلها

$$\ddot{y} = 2 e^x \cos(x)$$

If $y = x^2 + x \log(1-x)$ then $\ddot{y} = \dots$

حاصل ضرب

$$\dot{y} = 2x + (x)' \cdot \log(1-x) + (\log(1-x))'(x)$$

$$\dot{y} = 2x + (1) \log(1-x) + \frac{-1}{1-x} (x)$$

$$\dot{y} = 2x + \log(1-x) - \frac{x}{1-x}$$

$$\ddot{y} = 2 + \frac{-1}{1-x} - \frac{(x)'(1-x) - (1-x)'(x)}{(1-x)^2}$$

$$\ddot{y} = 2 - \frac{1}{1-x} - \frac{1(1-x) - (-1)(x)}{(1-x)^2}$$

$$\ddot{y} = 2 - \frac{1}{1-x} - \frac{1-x+x}{(1-x)^2} = 2 - \frac{1}{1-x} - \frac{1}{(1-x)^2}$$

ملاحظات مهمة جداً

(6)

$$\textcircled{1} (k \cdot f(x))' = k \cdot f'(x)$$

مشتقة (عدد × دالة)

العدد × مشتقة = الدالة

$$\text{Ex } (2 \ln(x))' = 2 \cdot \frac{1}{x} = \frac{2}{x}$$

$$(3 \sin(5x))' = 3 \cdot \cos(5x) (5) = 15 \cos(5x)$$

$$\textcircled{2} \left(\frac{f(x)}{k} \right)' = \frac{f'(x)}{k}$$

المقام ثابت

$$y = \frac{x}{2} \rightarrow \frac{dy}{dx} = \frac{1}{2}$$

$$y = \frac{e^x + 5x}{3} \rightarrow \frac{dy}{dx} = \frac{e^x + 5}{3}$$

$$\textcircled{3} (f(x) \pm g(x))' = f'(x) \pm g'(x)$$

المشتقة
تنتوزع في الجمع
والطرح فقط

$$y = x + \ln(x) \rightarrow \frac{dy}{dx} = 1 + \frac{1}{x}$$

$$y = \tan(x) - x^5 + 2x \rightarrow \frac{dy}{dx} = \sec^2(x) - 5x^4 + 2$$

$$\textcircled{4} (f(x) \cdot g(x))' = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

مشتقة الأولى × الثانية + الثانية × مشتقة الأولى

يمكن نقول

$$f'(x) \cdot g(x) + g'(x) \cdot f(x)$$

مشتقة الأولى × الثانية + مشتقة الثانية × الأولى

بشكل أفضل

$$y = x^2 \cdot \sin(x)$$

(7)

$$\frac{dy}{dx} = x^2 \cdot (\sin x)' + \sin(x) (x^2)$$

$$= x^2 \cos(x) + \sin(x) \cdot 2x$$

$$\frac{dy}{dx} = x^2 \cos(x) + 2x \sin(x)$$

$$\textcircled{5} \left(\frac{f(x)}{g(x)} \right)' = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

المقام × مشتقة البسط - البسط × مشتقة المقام
(المقام)²

أو يمكن
مشتقة البسط × المقام - البسط × مشتقة المقام
(المقام)²

نقطة هذا

$$y = \frac{1+x}{1-x}$$

$$\frac{dy}{dx} = \frac{(1+x)' \cdot (1-x) - (1-x)' \cdot (1+x)}{(1-x)^2}$$

$$= \frac{(1)(1-x) - (-1)(1+x)}{(1-x)^2} = \frac{1-x+1+x}{(1-x)^2}$$

$$\frac{dy}{dx} = \frac{2}{(1-x)^2}$$

④ $y = \log(a - 3x)$

$$y' = \frac{-3}{(a-3x)} \rightarrow y'' = \frac{-(-3)(1)(-3)}{(a-3x)^2} = \frac{-9}{(a-3x)^2} \quad (T)$$

يمكن تشتت كواله
كسرية

هكذا $y'' = \frac{(-3)'(a-3x) - (a-3x)'(-3)}{(a-3x)^2} = \frac{(0)(a-3x) - (-3)(-3)}{(a-3x)^2} = \frac{-9}{(a-3x)^2} \quad (T)$

⑤ $y = x^x$ (دالة أس دالة)

$$\frac{dy}{dx} = x^x (x \log x)'$$

$$= x^x ((1) \log(x) + \frac{1}{x} \cdot x)$$

$$\frac{dy}{dx} = x^x (\log(x) + 1) \quad (T)$$

⑥ $y = x^x - x$

$$y' = x^x (x \log x)' - (1) = x^x ((1) \log(x) + \frac{1}{x} \cdot x) = x^x (\log(x) + 1)$$

$$\therefore (x^x)' = x^x (\log(x) + 1)$$

حاصل ضرب والتبني

$$y'' = (x^x)' \cdot (\log(x) + 1) + (\log(x) + 1)' \cdot x^x$$

$$y'' = x^x (\log(x) + 1) \cdot (\log(x) + 1) + (\frac{1}{x} + 0) x^x$$

$$y'' = x^x (\log(x) + 1)^2 + \frac{1}{x} \cdot x^x \rightarrow (T) \quad x=1$$

$$y''(1) = (1)' (\log(1) + 1)^2 + \frac{1}{1} \cdot (1)' = 1 (1)^2 + 1(1) = 1 + 1 = 2$$

$$\log(1) = 0$$

$$(18) y = \log(\cos x)$$

$$\bar{y} = \frac{-\sin x}{\cos(x)} = -\tan(x)$$

$$\bar{\bar{y}} = -\sec^2(x) \rightarrow -(\sec x)^2$$

$$\bar{\bar{\bar{y}}} = -2(\sec(x))' \cdot \sec x \tan x$$

$$\bar{\bar{\bar{y}}} = -2 \sec^2(x) \tan x$$

$$(19) y = \sqrt{x+1} \rightarrow \text{نحوه الی قویا لانه مطلوب عدة مشتقات}$$

$$y = (x+1)^{\frac{1}{2}}$$

$$\bar{y} = \frac{1}{2} (x+1)^{-\frac{1}{2}} (1) = \frac{1}{2} (x+1)^{-\frac{1}{2}}$$

$$\bar{\bar{y}} = \frac{1}{2} \left(-\frac{1}{2}\right) (x+1)^{-\frac{3}{2}} (1) = -\frac{1}{4} (x+1)^{-\frac{3}{2}}$$

$$\bar{\bar{\bar{y}}} = -\frac{1}{4} \left(-\frac{3}{2}\right) (x+1)^{-\frac{5}{2}} = \frac{3}{8} (x+1)^{-\frac{5}{2}}$$

$$(20) y = \frac{1}{\sqrt{1-x}}$$

نبت $\frac{1}{x^n} = x^{-n}$

$$y = \frac{1}{(1-x)^{\frac{1}{2}}} \uparrow = (1-x)^{-\frac{1}{2}}$$

$$\bar{y} = -\frac{1}{2} (1-x)^{-\frac{3}{2}} (-1) = \frac{1}{2} (1-x)^{-\frac{3}{2}}$$

$$\bar{\bar{y}} = \frac{1}{2} \left(-\frac{3}{2}\right) (1-x)^{-\frac{5}{2}} (-1) = \frac{3}{4} (1-x)^{-\frac{5}{2}}$$

$$\bar{\bar{\bar{y}}} = \frac{3}{4} \left(-\frac{5}{2}\right) (1-x)^{-\frac{7}{2}} (-1) = \frac{15}{8} (1-x)^{-\frac{7}{2}}$$

$$(21) y = \log(\sin x)$$

$$\bar{y} = \frac{\cos x}{\sin x}$$

$$\bar{y} = \cot(x)$$

$$\bar{\bar{y}} = -\operatorname{cosec}^2(x) \rightarrow -(\operatorname{cosec} x)^2$$

$$\bar{\bar{\bar{y}}} = -2(\operatorname{cosec} x)' (-\operatorname{cosec}(x) \cot(x))$$

$$\bar{\bar{\bar{y}}} = 2 \operatorname{cosec}^2(x) \cot(x)$$